

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE211	COMPUTER ORGANIZATION AND DESIGN	3	1	0	4	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50	Exam Category: 11: Mid Term Exam: All MCQ – End Term Exam: All MCQ				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: define the basics of organization and architecture of a digital computer

CO2 :: explain the concept of Instruction sets, microprocessor design and addressing modes

CO3 :: examine the input-output unit and its processing in a digital computer

CO4 :: analyze Verilog HDL, in relation of Register Transfer to model, simulate, and verify digital systems

CO5 :: articulate the performance of various computer memory systems and analyze parallel processing

CO6 :: analyze modern computing architectures, including GPU architecture and understand latest trends in computer architecture

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	COMPUTER SYSTEM ARCHITECTURE	M. MORRIS MANO, RAJIB MALL	PEARSON
	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	COMPUTER ORGANIZATION AND ARCHITECTURE	WILLIAM STALLINGS	Pearson Education India
R-2	DIGITAL SYSTEMS DESIGN USING VERILOG	CHARLES ROTH, LIZY K. JOHN, BYEONG KIL LEE	Cengage India Private Limited

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

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Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Computer Organization (computer registers)	T-1		Introduction to basic computer organization and show how its operation can be specified with register transfer statement.	Students will understand about internal registers, control structure and instruction . ii. Understanding of Common bus system	Discussion with examples, peer learning, PPT demonstration with images	
		Computer Organization (common bus system)	T-1		Introduction to basic computer organization and show how its operation can be specified with register transfer statement.	Students will understand about internal registers, control structure and instruction . ii. Understanding of Common bus system	Discussion with examples, peer learning, PPT demonstration with images	
		Computer Organization (common bus system)	T-1		Introduction to basic computer organization and show how its operation can be specified with register transfer statement.	Students will understand about internal registers, control structure and instruction . ii. Understanding of Common bus system	Discussion with examples, peer learning, PPT demonstration with images	
	Lecture 2	Computer Organization (computer instructions)	T-1		Explain the basics of instruction execution on a computer.	Explain the basics of instruction execution on a computer.	Peer learning, demonstration with images	
		Computer Organization (timing and control)	T-1		Explain the basics of instruction execution on a computer.	Explain the basics of instruction execution on a computer.	Peer learning, demonstration with images	
	Lecture 3	Computer Organization (instruction cycle)	T-1		Discussion on instruction cycle, memory reference instructions and interrupt cycle	Students will identify and analyze the design and function of the basic instruction execution elements of a modern processor	Discussion thorough knowledge and understanding of key concepts	

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Week 2	Lecture 4	Computer Organization (memory reference instructions)	T-1 R-1		Discussion on memory reference instructions and interrupt cycle	`Students will identify and analyze the design and function of the basic instruction execution elements of a modern processor	Discussion thorough knowledge and understanding of key concepts	
		Computer Organization (input-output and interrupt)	T-1 R-1		Discussion on memory reference instructions and interrupt cycle	`Students will identify and analyze the design and function of the basic instruction execution elements of a modern processor	Discussion thorough knowledge and understanding of key concepts	
	Lecture 5	Computer Organization(lmc (simulation software))	T-1 R-1		Introduces the Little Man Computer (LMC) simulation software to help students understand basic computer architecture and instruction execution through a simplified virtual model.	Students will be able to design, simulate, and analyze simple assembly programs using the LMC environment.	Problem-based learning through sample programs Classroom discussions and doubt clarification	
	Lecture 6	Central Processing Unit (General Register Organization)	T-1		Introduction to various data processing operations performed in CPU and storage device that store information	i) Understand the organization and architecture of CPU. ii) To understand stack based processor organization. ii. Instruction set of a stack organized processor	Discussion with examples, peer learning, demonstration with images	
		Central Processing Unit (Stack Organization)	T-1		Introduction to various data processing operations performed in CPU and storage device that store information	i) Understand the organization and architecture of CPU. ii) To understand stack based processor organization. ii. Instruction set of a stack organized processor	Discussion with examples, peer learning, demonstration with images	

Week 3	Lecture 7	Central Processing Unit (Reduced instruction set computer)	T-1		CISC and RISC variable length instructions in multiple cycles	Students will understand the CISC and RISC Architecture	peer learning, demonstration with images	
		Central Processing Unit (Complex instruction set computer)	T-1		CISC and RISC variable length instructions in multiple cycles	Students will understand the CISC and RISC Architecture	peer learning, demonstration with images	
	Lecture 8	Central Processing Unit (instruction formats)	T-1		It explains different instruction formats used in computer architecture, focusing on the structure of instructions, field divisions (opcode, operand, addressing mode), and their role in efficient instruction execution.	Students will be able to identify, analyze, and differentiate various instruction formats used in processors.	Problem-solving exercises Interactive questioning and discussion	
	Lecture 9	Central Processing Unit (Data Transfer Schemes)	T-1		Describing about different modes of transfer and categories of computer instruction including program control instructions and data transfer schemes	Students will learn about computer instruction and data transfer instruction	Demonstration with images, real examples, peer learning	
Week 4	Lecture 10	Central Processing Unit (Program Control and Interrupts)	T-1		Describes program control mechanisms and interrupt handling in computer systems, explaining control flow instructions, interrupt types, interrupt processing, and their importance in efficient CPU operation.	students will be able to explain program control instructions and analyze interrupt handling mechanisms in modern processors.	Demonstration with images, real examples, peer learning	
	Lecture 11	Central Processing Unit (Processor status word)	T-1		Describing about different modes of transfer and categories of computer instruction including program control instructions and PSW	Students will learn about computer instruction and data transfer instruction and PSW	Demonstration with images, real examples, peer learning	

Week 4	Lecture 12	Central Processing Unit (Addressing Modes)	T-1		introduces different addressing modes used in computer architecture, explaining how operands are specified, accessed, and utilized during instruction execution.	tudents will be able to identify, compare, and apply various addressing modes in instruction execution.	Discussion with examples, peer learning, demonstration with images	
Week 5	Lecture 13	Input-Output Organization (Peripheral Devices)	T-1		Introduction to peripheral devices used in computer systems, explaining their types, functions, and interaction with the CPU and memory through input, output, and storage operations.	Students will be able to identify different peripheral devices and explain their roles in a computer system.	Flipped classroom, Demonstration with images, PPTs	
	Lecture 14	Input-Output Organization (Priority interrupt)	T-1		Introduction on priority interrupt systems, focusing on interrupt priority levels, arbitration methods, and their role in managing multiple interrupt requests efficiently.	Students will be able to analyze priority interrupt mechanisms and explain how processors handle multiple simultaneous interrupt requests.	Peer to peer learning	
	Lecture 15				Test 1			
Week 6	Lecture 16	Input-Output Organization (Input output interface)	T-1		Describes the concept of input–output (I/O) interfaces, explaining how peripheral devices communicate with the CPU, the role of interface circuits, and data transfer methods between devices and the system.	Students will be able to explain the functioning of I/O interfaces and analyze different data transfer techniques used in computer systems.	Flipped classroom, Demonstration with images, PPTs	
	Lecture 17	Input-Output Organization (Direct memory access transfer)	T-1		Discussion on DMA	Students will understand the concept of DMA	Peer to Peer Learning	
	Lecture 18	Input-Output Organization (Input/Output processor.)	T-1		Discuss its architecture, functions, and role in managing I/O operations independently of the main CPU to improve system performance.	Students will be able to describe the working of an I/O processor and analyze its advantages in handling input–output operations.	Flipped classroom, Demonstration with images, PPTs	

Week 7	Lecture 19	Input-Output Organization (modes of data transfer)	T-1		Introduction to input output subsystem of a computer	Students will understand the mode of communication between the central system and out side environment UART	Demonstration with PPT, videos	
		Input-Output Organization (UART)	T-1		Introduction to input output subsystem of a computer	Students will understand the mode of communication between the central system and out side environment UART	Demonstration with PPT, videos	
		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	Introduction to HDLs in Digital Systems(History)	R-2		Describes Hardware Description Languages (HDLs) in digital systems, focusing on their history, evolution, and significance in modern digital design and verification processes.	Students will be able to explain the development of HDLs and analyze their role in contemporary digital system design.	Demonstration with PPT, videos	
		Introduction to HDLs in Digital Systems(Evolution)	R-2		Describes Hardware Description Languages (HDLs) in digital systems, focusing on their history, evolution, and significance in modern digital design and verification processes.	Students will be able to explain the development of HDLs and analyze their role in contemporary digital system design.	Demonstration with PPT, videos	
	Lecture 23	Introduction to HDLs in Digital Systems(Typical HDL-based Design Flow (Design, Simulation, Synthesis, Verification))	R-2		Introduction of complete HDL-based design flow, including design entry, simulation, synthesis, and verification stages, highlighting their roles in developing reliable digital systems.	Students will be able to describe each stage of the HDL design flow and apply basic verification techniques to validate digital designs.	Discussion and Demonstration with PPT, videos and images	

Week 8	Lecture 24	Introduction to HDLs in Digital Systems(Comparison with Software Programming Languages)	R-2		Describe Hardware Description Languages (HDLs) with software programming languages, focusing on Verilog, to highlight differences in execution, concurrency, timing, and hardware modeling concepts.	Students will be able to differentiate between HDL and software programming concepts and explain Verilog's role in digital hardware design.	Classroom Discussion and problem solving exercises.	
		Introduction to HDLs in Digital Systems(Verilog)	R-2		Describe Hardware Description Languages (HDLs) with software programming languages, focusing on Verilog, to highlight differences in execution, concurrency, timing, and hardware modeling concepts.	Students will be able to differentiate between HDL and software programming concepts and explain Verilog's role in digital hardware design.	Classroom Discussion and problem solving exercises.	
Week 9	Lecture 25	Introduction to HDLs in Digital Systems(Register Transfer and Micro Operation)	T-1		Introduction to Register Transfer and Micro Operation	Students will understand about registers	Discussion with examples, peer learning	
	Lecture 26	Introduction to HDLs in Digital Systems(Register Transfer Language)	T-1		Introduction to Register Transfer and Micro Operation	Students will understand about RTL and also about Bus and Memory Transfer	Discussion with examples, peer learning, PPT demonstration with images	
		Introduction to HDLs in Digital Systems(Bus and Memory Transfer)	T-1		Introduction to Register Transfer and Micro Operation	Students will understand about RTL and also about Bus and Memory Transfer	Discussion with examples, peer learning, PPT demonstration with images	
	Lecture 27	Introduction to HDLs in Digital Systems(Shift Register)	T-1		Introduction to Register Transfer and Micro Operation	Students will understand about arithmetic logic shift unit	Discussion with examples, peer learning, PPT demonstration with images	
		Introduction to HDLs in Digital Systems(Micro operations (Arithmetic, logic, shift), Addition, Subtraction, Multiplication Algorithm)	T-1		Introduction to Register Transfer and Micro Operation	Students will understand about arithmetic logic shift unit	Discussion with examples, peer learning, PPT demonstration with images	
Week 10	Lecture 28	Introduction to HDLs in Digital Systems(Booth Multiplication)	T-1		Working of multiplication in a digital computer	Students will be able to understand working of multiplication in a digital computer	Practice with examples	Working on different examples

Week 10	Lecture 29	Memory Unit(Memory hierarchy)	T-1		Discussion on memory unit	Students will understand the concept of computer memory	Working on LMC	Taking examples of association from web indexing, etc.
		Memory Unit(main memory)	T-1		Discussion on memory unit	Students will understand the concept of computer memory	Working on LMC	Taking examples of association from web indexing, etc.
	Lecture 30	Memory Unit(Virtual memory)	T-1		Students will understand the concept of computer memory	Students will understand the concept of computer memory	Flipped classroom, Demonstration with images, PPTs	
		Memory Unit(auxiliary memory)	T-1		Students will understand the concept of computer memory	Students will understand the concept of computer memory	Flipped classroom, Demonstration with images, PPTs	
Week 11	Lecture 31	Memory Unit(Cache memory)	T-1		Discussion on memory unit	`Students will understand the concept of computer memory	Peer to peer learning	
		Memory Unit(Writing into Cache concept)	T-1		Discussion on memory unit	`Students will understand the concept of computer memory	Peer to peer learning	
		Memory Unit(Mapping Techniques)	T-1		Discussion on memory unit	`Students will understand the concept of computer memory	Peer to peer learning	
	Lecture 32				Test 2			
	Lecture 33	Memory Unit(Introduction to Parallel Processing)	T-1		Detailed discussion on instruction pipelining in computer.	Introduction to pipelining and its types	Peer Learning	
		Memory Unit(Pipelining)	T-1		Detailed discussion on instruction pipelining in computer.	Introduction to pipelining and its types	Peer Learning	
Week 12	Lecture 34	Memory Unit (Characteristics of Multiprocessors)	T-1		Introduction to parallel processing and multiprocessors	Students will learn about multiprocessors system	Discussions with images and diagrams	

Week 12	Lecture 35	Memory Unit (Interconnection Structures)	T-1		Introduction to parallel processing and Various multiprocessor interconnection structures	Students will learn about Time-shared, Multiport memory, crossbar switch, hypercube interconnection and Amdahl's law	Discussions with images and diagrams	
		Memory Unit(Parallel Processing)	T-1		Introduction to parallel processing and Various multiprocessor interconnection structures	Students will learn about Time-shared, Multiport memory, crossbar switch, hypercube interconnection and Amdahl's law	Discussions with images and diagrams	
	Lecture 36	Introduction to GPU Architecture(Nvidia Case Study)	T-1 R-2		Introduction to GPU architecture with a focus on NVIDIA as a case study, explaining core components, parallel processing concepts, memory hierarchy, and the role of GPUs in high-performance computing and AI applications.	Students will be able to explain GPU architecture concepts and analyze NVIDIA GPU design for parallel computing applications.	Interactive discussion and problem solving	
Week 13	Lecture 37	Introduction to GPU Architecture(Introduction to Supercomputer)	T-1 R-2		Describes supercomputers, explaining their architecture, performance metrics, classifications, and applications in scientific research, weather forecasting, and artificial intelligence.	Students will be able to describe supercomputer architecture and explain their role in solving complex computational problems.	Discussion through images, ppts and videos	
	Lecture 38	Introduction to GPU Architecture(Introduction to Qubits and Quantum Computing)	T-1 R-2		Introduction to fundamentals of quantum computing, focusing on qubits, superposition, entanglement, and basic quantum operations, highlighting their potential impact on future computing technologies.	Students will be able to explain the concept of qubits and describe basic principles of quantum computing.	Interactive discussions and video demonstrations	

Week 13	Lecture 39	Introduction to GPU Architecture(Latest Technology and Trends in Computer Architecture)	T-1 R-1		Discussion on new nonvolatile memory technology and discussion on various processors	Students will learn about new processor and new memory technology with there architectures	Peer Learning	
		Introduction to GPU Architecture(Next Generation Processors Architecture)	T-1 R-1		Discussion on new nonvolatile memory technology and discussion on various processors	Students will learn about new processor and new memory technology with there architectures	Peer Learning	
Week 14	Lecture 40	Introduction to GPU Architecture (Microarchitecture)	T-1 R-1		Latest processor architecture study is done and Discussion on new nonvolatile memory technology and discussion on various processors	Latest processor architecture study is done and Students will learn about new processor and new memory technology with there architectures	Demonstrations, Discussions and Peer Learning	
		Introduction to GPU Architecture(Latest Processor for Smartphone or Tablet and Desktop)	T-1 R-1		Latest processor architecture study is done and Discussion on new nonvolatile memory technology and discussion on various processors	Latest processor architecture study is done and Students will learn about new processor and new memory technology with there architectures	Demonstrations, Discussions and Peer Learning	
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Recoup CA		

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Test 1	50	CO1, CO2
Test 2	50	CO3, CO4, CO5

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Test 1	To assess the students towards their understanding of topics taught	MCQ based question on unit 1,2. Questions must follow analytical and scenario based approach and also must follow revised blooms taxonomy	Individual	Online	30	4 / 5
Test 2	To assess the students towards their understanding of topics taught	MCQ based question on unit 3,4 and 5. Questions must follow analytical and scenario based approach and must follow blooms taxonomy	Individual	Online	30	10 / 11
Recoup CA	subjective test based on the syllabus	Additional chance of CA	Individual	Online	30	8 / 10

Plan for Tutorial: (Please do not use these time slots for syllabus coverage)

Tutorial No.	Lecture Topic	Type of pedagogical tool(s) planned (case analysis,problem solving test,role play,business game etc)
Tutorial1	Problem Solving on Logic Gates	Problem Solving
Tutorial2	Problem solving on Control Timing Signals and Instruction Cycle	Problem Solving
Tutorial3	Problems based on Memory Reference Instructions, Input-Output and Interrupts	Problem Solving
Tutorial4	Problem solving on Data Transfer and Data Manipulation	Problem Solving
Tutorial5	Problem solving on Addressing Modes	Problem Solving
Tutorial6	Problem solving on Direct Memory Access	Problem Solving
Tutorial7	Problem Solving on Priority Interrupts	Problem Solving
After Mid-Term		
Tutorial8	Problem solving on HDL on digital system	Problem Solving

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Tutorial9	Problem solving on Arithmetic and Micro Shift Operations	Problem Solving
Tutorial10	Problem solving on Register Transfer Language and Bus	Problem Solving
Tutorial11	Problem solving on Cache Memory and Virtual memory	Problem Solving
Tutorial12	Problem solving on Performance of Processors with Pipelining	Problem Solving
Tutorial13	Problem solving on Digital Computer Arithmetic	Problem Solving
Tutorial14	Problem solving on Flynn's Classification	Problem Solving

